

Zahir Ali

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Education

University of California, San Diego

Sep 2024 - June 2027

B.S. Cognitive Science: Machine Learning & Neural Computation, Minor in Computer Science

Coursework: Machine Learning (ML) Algorithms, Supervised & Unsupervised Machine Learning, Large Language Models (LLM), Advanced Data Structures, Advanced Python, Data Science in Practice, Modeling & Data Analysis, Computer Architecture, Software Tools, Brain-Computer Interfaces

Leadership: VP of UCSD Tech Collaborative | VP of Alpha Lambda Mu | VP of Bengali Student Association

Experience

Atmosense, Inc. (NSF SBIR Phase I Awardee)

San Diego, CA

Embedded Systems Software Engineering Intern

Sep 2025 – Present

- Engineered signal-processing and calibration algorithms in Python and embedded C, collaborating with hardware engineers to enable ppb-level gas detection of formaldehyde and methane for consumer IoT integration.
- Optimized linear-regression algorithm for microcontroller firmware, reducing memory footprint by 40% while maintaining 95%+ calibration accuracy under real-time constraints.
- Co-authoring technical documentation for NSF SBIR Phase II proposal (Q1 2026); designed and presented 5+ Power BI dashboards visualizing sensor calibration data for executive decision-making.

UC San Diego School of Physical Sciences

San Diego, CA

Undergraduate Machine Learning Researcher

June 2025 – July 2025

- Developed TD3-LSTM reinforcement learning policy for decentralized underwater swarm navigation, enabling autonomous coordination of 10+ robots without centralized control in physics-accurate simulations.
- Architected custom Gymnasium environment with goal-reaching tasks and reward shaping, reducing training convergence time by 30% compared to baseline configurations.
- Evaluated RL algorithms (TD3, SAC, PPO) across 5–50 robot swarm configurations; identified TD3-LSTM as optimal for scalable multi-agent coordination.

Projects

Theology AI Evaluation Benchmark | *Python, LangChain, NLP, lm-eval-harness*

- Led development within 12-person team to build the first Theology AI leaderboard with 1,000+ curated prompts and custom evaluation metrics for GPT and Claude models.
- Implemented automated multiagent evaluation pipeline using LangChain for systematic LLM assessment on Islamic ethics and theological knowledge.

San Diego Injury Prevention Program Website | *React, TypeScript, Tailwind CSS, Firebase*

- Designed and developed a responsive public health website for the San Diego Injury Prevention Program, focused on accessibility, community outreach, and resource navigation.
- Integrated Firebase backend services for content management and scalable deployment.

Startup Success Prediction | *Python, SQL, R, Tableau*

- Led 5-person team analyzing 70,000+ Crunchbase records; built predictive models identifying acquisition vs. failure indicators with 78% classification accuracy.
- Performed data cleaning, feature engineering, and statistical analysis; delivered findings via Tableau dashboards and Jupyter documentation.

Technical Skills

Languages: Python, Java, C/C++ (Embedded), JavaScript, SQL, R, HTML/CSS, ARM Assembly

Tools & Frameworks: PyTorch, TensorFlow, scikit-learn, OpenAI API, LangChain, RAG, Stable-Baselines3, Gymnasium, NumPy, Pandas, React, Streamlit, PostgreSQL, SQLite, MongoDB, Firebase, AWS, GraphQL, Power BI, Tableau, Git, Linux/Unix, Docker, Figma, JIRA, Anaconda